

J8L Application Guide

Ver. 1.1 For J8L firmware ver. 0.3.8

Date. 27, 09, 2024

Supported Command List

Command	Description	Arguments	Return Value	Notes	Functional Verification
ToF Commands					
settofconf	Turns on the ToF sensor	N/A	OK/NG Note 1		OK
settofpwr	Turns off the ToF sensor	0	OK/NG Note 1	Interrupts are disabled, but data continues	OK
Set Data Commands					
setlcdinvl	Set left LCD panel inversion	0-3 (Inversion directions)	OK/NG Note 1	0: Left/Up (default) 1: Right/Up 2: Left/Down 3: Right/Down	OK
setlcdinvr	Set right LCD panel inversion	0-3 (Inversion directions)	OK/NG Note 1	0: Left/Up (default) 1: Right/Up 2: Left/Down 3: Right/Down	OK
setlcdluxl	Set left LCD panel luminance	1000-5000 (Multiples of 100)	OK/NG Note 1	Luminance must be a multiple of 100	OK
setlcdluxr	Set right LCD panel luminance	1000-5000 (Multiples of 100)	OK/NG Note 1	Luminance must be a multiple of 100	OK
setlcdhorl	Set left LCD horizontal orbit	+: 000-016, -: 100-116	OK/NG Note 1	0 = positive orbit 1 = negative orbit	OK
setlcdhorr	Set right LCD horizontal orbit	+: 000-016, -: 100-116	OK/NG Note 1	0 = positive orbit 1 = negative orbit	OK
setlcdvorl	Set left LCD vertical orbit	+: 000-016, -: 100-116	OK/NG Note 1	0 = positive orbit 1 = negative orbit	OK
setlcdvorr	Set right LCD vertical orbit	+: 000-016, -: 100-116	OK/NG Note 1	0 = positive orbit 1 = negative orbit	OK
setlcddefault	Set all LCD setting to default value	N/A	OK/Error	Resets all settings stored in flash	OK
setlcdmode	Set display mode (60/120Hz, 2D/3D)	0: 60Hz/2D 1: 120Hz/2D 2: 60Hz/3D 3: 120Hz/3D	OK/NG Note 1	Wait for device stability before switching modes	OK

setlcddevice1	Set left panel serial number	String (< 18 characters)	OK/Error	Serial number must be 18 characters or less	OK
setlcddevice2	Set right panel serial number	String (< 18 characters)	OK/Error	Serial number must be 18 characters or less	OK
Get Data Commands					
getfirmver	Get the firmware version	N/A	Firmware version		OK
getbrgfw	Get the firmware version of the bridge	N/A	Firmware version		OK
getlcdvalue	Get current LCD settings	N/A	LCD setting	Shows the current LCD settings	OK
Flash Commands					
flashr	Read LCD settings from flash	N/A	OK/Error	Replaces current LCD settings with those from flash	OK
flashw	Write current LCD settings to flash	N/A	OK/Error	Saves current LCD settings to flash	OK
Special Commands					
enterbootloader	Enter bootloader mode	N/A	Not Implemented	Immediately switches the system to DFU mode	OK

Note 1: If the message returns "NG", an error code will follow: 0 = Invalid Command, 1 = Invalid Parameter, 2 = I/O Error, 3 = Command Not Allowed (Incorrect System State)

Button Event List

Event Type	Description	Action/Effect	Return Value	Example	Functional Verification
volume	Triggered when BTN_VOL_PLUS or BTN_VOL_MINUS is pressed	Adjusts the speaker volume	+1/-1 ^{Note 1}	#volume +1@	OK
lcdmode	Triggered when BTN_LCD_MODE is pressed	Displays the current LCD mode after the mode is changed	0, 2 ^{Note 1,2}	#lcdmode 2@	OK

Note 1: The returned CDC events are prefixed with the '#' symbol and end with the '@' symbol.

Note 2: For lcdmode, the value alternates between 0 (60Hz/2D) and 2 (60Hz/3D), as 120Hz modes have been removed.

